Algorithm	Description
Point cloud segmentation	Used to separate the lidar data into individual objects, such as cars, pedestrians, and buildings. This enables the car to identify and track objects in its environment.
Object detection and classification	Used to identify the type of objects in the environment, such as cars, pedestrians, and bicycles. These algorithms use a combination of geometric and appearance-based features to identify objects in the lidar data.
Simultaneous localisation and mapping (SLAM)	Simultaneous localisation and mapping (SLAM) algorithms are used to create a map of the environment in real-time. This involves using the lidar data to create a 3D map of the environment, as well as tracking the vehicle's position and orientation within that map.
Object tracking	Used to track the movement of objects in the environment, such as other vehicles and pedestrians, using a combination of prediction and filtering techniques to track the object's movement.
Sensor fusion	Used to integrate lidar data with other sensor data, such as camera and radar data to improve its accuracy and reliability in identifying and tracking objects in the environment.
Localisation	Localisation algorithms are used to determine the car's precise location in the world using data from a variety of sources, including GPS, lidar sensors, and cameras, to pinpoint the car's location.

Table 2: Algorithms used in lidar processing

Algorithm	Description
A* Search algorithm	This searches for the shortest path between start and end points, while taking into account the distance to the goal and obstacles in the environment.
Probabilistic roadmap (PRM)	Generates a roadmap of the environment by randomly selecting points and connecting them with paths. Searches for the shortest path between start and end points using the roadmap.
Rapidly-exploring random trees (RRT)	Constructs a tree of possible paths from the start to the end point by exploring the environment by randomly selecting points and extending the tree towards them. Searches for the shortest path between start and end points using the tree.
Dynamic programming	Finds the shortest path between the start and end points by recursively solving sub-problems.

Table 3: Path planning algorithms

Algorithm	Description
Adaptive cruise control (ACC)	Enables the vehicle to maintain a safe distance from the vehicle in front of it by adjusting the speed of the car using sensors such as radar or lidar.
Lane keeping assistance (LKA)	Helps the vehicle stay within its lane by detecting lane markings on the road using cameras or other sensors and applying steering corrections.
Automatic emergency braking (AEB)	Automatically applies the brakes in emergency situations using sensors such as radar or cameras to detect obstacles in the vehicle's path and avoid a collision.
Path planning	Determines the best path for the vehicle to follow through its environment using data from sensors such as lidar and cameras to create a map of the environment and determine the safest and most efficient path for the vehicle to follow.
Trajectory planning	Determines the vehicle's movement and speed along the path, considering factors such as the vehicle's speed, steering angle, and braking to plan a safe and efficient trajectory along the path.

Table 4: Control algorithms

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